ELEC 872 Week 8 Lab

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1. Introduction

In this lab, we explore the task of facial expression recognition (FER) using the FER+ [1] and FER2013 [3] datasets. Our baseline is a ResNet18 [4] model initialized from the ImageNet-1K checkpoint available from PyTorch. We show the results in Table 1 on FER+ after 25 epochs under the name `baseline`.

2. Input Size

Our first experiment looks to see the effect of the input image size on the expression recognition performance. The origninal images from the dataset are 48×48 px in dimension, but are rescaled to 128×128 px. This introduces some extra computation to produce the output, and so we test to see if this extra computation is worth it. We retrain the ResNet18 [4] network without resizing the inputs with all else being identical to the baseline. The results are shown in Table 1 under the name `48px input`.

The results show that artificially increasing the input image resolution also increases the performance. Indeed, the overall metrics all drop by about 3% compared to the baseline method. This can seem a bit counterintuitive at first since interpolation in the resizing adds no extra information, but there are a few reasons why this might happen:

- *More computation*: With a larger image, we obtain more activations and hence more features for the model to process and use in its classification.
- *Compatibility with pre-training*: From the ImageNet-1K checkpoint initialization, the learned filters inside the model are used to seeing inputs of size 224×224 px. The larger 128×128 px images in the baseline are much closer to the pre-training image resolution than to the tested 48×48 px, making the previously learned filters more compatible and transferrable to the images in the baseline.

3. Inproving SoTA Performance

One of the state-of-the-art works in facial expression recognition is EfficientFER [6], which claims to achieve 82.47% accuracy on FER2013 [3]. The authors themselves claim to have used extra training data by generating 11,000 synthetic face images using Imagen 3 [5]. The extra training data was not released. The method itself uses EfficientNetv2-S [8] as the backbone and is trained for 50 epochs. The results of our reproduction attempt is shown near the bottom of Table 1. Note that the tested FER2013 dataset is not directly comparable with the FER+ dataset used for our other experiments. Most importantly, we were not able to reproduce the claimed 82.47% accuracy, instead achieving a mere 70.9% testing accuracy. One major cause could be the lack of the synthetically generated data that was mentioned in the original paper [6].

The implementation already uses many techniques to improve the performance of the network, including using the focal loss and keeping an exponential moving average (EMA) of the model weights over the course of training. In our experiments, we propose to add a learning rate schedule to make the network converge better. We use the OneCycleLR scheduler[1] available in PyTorch which first warms up the learning rate up to a maximum value of 10^{-3} roughly 30% into the training, then decays the learning rate for the rest of training according to a cosine schedule. The results of this experiment are shown in Table 1. Overall, we see consistent improvement over the original EfficientFER model with the macro F_1 score increasing by about 1%. Scheduling the learning rate ensures stability of the initialization at the beginning of training and fine weight adjustments near the end.

It is worth noting that we originally tried adding CutMix [9] and MixUp [10] augmentations to the training, but ultimately the macro F_1 score dropped by about 2%. This could be because facial expressions are more subtle than the entirely-different-object classes that these augmentations were originally meant for. MixUp blurs the image, making smaller distinctions harder to tell, while CutMix may copy and paste non-discriminable regions (e.g. ears, cheek) from other images, effectively acting as noise in the prediction.

[1] https://docs.pytorch.org/docs/2.15/generated/torch.optim.lr_scheduler.OneCycleLR.html

Table 1. Performance of the tested methods and ablations on the FER+ and FER2013 datasets. All metrics are reported in percentages where higher is better. Per-expression F_1 scores are reported along with accuracy and F_1 scores across each class. The choice of ImageNet-1K pre-training is essential for the high performance of the ResNet-18 architecture, in addition to using upscaled 128×128 px inputs.

Method	Per-Expression F_1								Overall Metrics		
	Neutral	Happy	Surprise	Sad	Angry	Disgust	Fear	Contempt	Accuracy	Macro F_1	Weighted F_1
FER+ Dataset											
baseline	**86.8**	**94.8**	**88.3**	**71.5**	**82.4**	**43.5**	**65.5**	**54.5**	**85.8**	**73.4**	**85.7**
48px input	84.7	91.7	85.8	68.2	80.1	39.1	55.6	43.6	82.7	68.6	82.8
no pretrain	84.3	92.6	84.5	65.4	80.6	43.2	55.2	34.5	82.5	67.5	82.4
AlexNet	84.3	92.3	84.7	67.0	79.0	50.0	57.9	37.0	82.5	69.0	82.5
face recognition	84.6	91.8	85.5	66.9	77.3	43.1	56.7	31.4	82.1	67.2	82.3
FER2013 Dataset											
EfficientFER	67.0	89.1	81.9	58.1	**64.7**	**77.8**	55.5	—	70.9	70.6	70.8
+ LR scheduler	**67.5**	**89.5**	**84.2**	**61.0**	64.5	75.6	**58.1**	—	**72.0**	**71.5**	**72.1**

4. Weight Initialization

This section explores the effect of different weight initializations and architectures on the performance on the FER+ dataset. We compare our ablations to the baseline which has weights initialized from pre-training on ImageNet-1K.

4.1. Random Initialization

We first attempt to train a version of the baseline ResNet18 model except without the initialization from the ImageNet-1K checkpoint. The results are shown as `no pretrain` in Table 1.

We see that our model without ImageNet pretraining performs worse than the baseline model, dropping significantly in the overall macro F_1 score. Despite not being explicitly trained on human emotions, the existing pre-training gives the model some useful priors, such as detecting edges and curves that can be transferred to the task of recognizing peoples' facial expressions.

4.2. AlexNet

We also experiment with the AlexNet [7] model as an alternative architecture for facial expression recognition. We see in Table 1 that rather unsurprisingly, it is beaten by the deeper and more recent ResNet18 model. AlexNet has 8 layers in total versus the 18 in ResNet18, meaning that the features learned by AlexNet are shallower and less complex, with a lower receptive field in the convolution layers. In terms of performance we can see that the more complex features learned by ResNet18 are necessary for good performance on the FER+ dataset.

4.3. AlexNet vs. ResNet18 Analysis

In terms of the individual performance on FER+, we can see that the ResNet18 baseline outperforms AlexNet on all categories in Table 1. During training, AlexNet had a training loss of 1.279, a validation loss of 1.345, a validation accuracy of 82.89%, and a macro F_1 of 70.75%. For the ResNet18 baseline, the training loss is 1.060, the validation loss is 1.272, the validation accuracy was 86.84%, and the validation macro F_1 was 76.08%. Both models trained very well without much overfitting just by seeing the difference between train and validation losses. Though, we see that ResNet18 performs much better in the validation macro F_1, being over 5% more than the ResNet18 baseline.

4.4. Pre-trained face recognition

We use a pre-trained ResNet18 checkpoint trained on VGGFace2 [2], a face recognition dataset with 3.3 million face images. This is used to initialize the ResNet18 model, which has the exact same architecture as the baseline. The results are shown in Table 1. The results are quite interesting: even though this checkpoint was trained on images of peoples' faces, the final set of metrics is worse than the baseline ImageNet initialization. This suggests that the features learned by the face recognition model are not transferrable to the task of facial expression recognition. Indeed, the goals of the two tasks are quite orthogonal: in face recognition, our goal is to learn the identity of the subject while discarding all information about their facial expression. On the other hand, in facial expression recognition, the goal is to learn the facial expression represented in the image while ignoring the subject of the identity.

5. Empirical Study

We conduct some empirical study on our proprietary Stuart-Yujong dataset. This is an acted facial expression recognition dataset consisting of two students with supposedly graduate levels of expertise. We control for the levels of

physiological signals in all subjects by constraining them to a maximum of 4 hours of sleep and having them run under the assumption that they have a lab due at 4:30pm. We use the baseline ResNet18 model trained on the FER+ dataset from Section 1 as it had the highest performance across all of our experiments. Example images and their predicted and ground truth emotion labels are shown in Figures 1 and 2.

In general, the model performs very well on the real-world examples in the Stuart-Yujong dataset. We see that the model has some bias towards the neutral emotion class. On the randomly sampled Stuart examples, we can see that more exaggerated and acted expressions are needed for the model to recognize the correct emotion. We can see this in juxtaposition to the Yujong images, where most of the images are predicted as neutral with high confidence, even though we can clearly tell one of the emotions from the others.

6. Conclusion

From our evaluations on the FER+ dataset, we can see that our baseline model for facial expression recognition, ResNet18, performs the best out of the methods tested. Specifically, the initialization from the ImageNet-1K pre-training stands out as the most effective in comparison to the random initialization as well as initialization from the pre-trained face recognition model. We have also seen that the choice of architecture and setup matters: using a shallower network like AlexNet does not learn the complex features required for the best performance on facial expression recognition tasks, and that images closely matching the pre-training image resolution of 224×224 px are essential for the transferability the learned image filters.

On the FER2013 dataset, we can see that synthetically generated images may be much needed to match the claimed accuracy for large models like EfficientFER [6], and that learning rate scheduling can better improve the models' convergence.

7. Generative AI Disclosure

Other than the AI usage from the given lab starter template, Claude was used to generate a helper script to split part of the training set of FER2013 into a validation set, as well as modify the existing focal loss in EfficientFER to work with CutMix / MixUp augmentations. All other work was done without AI assistance.

References

[1] Emad Barsoum, Cha Zhang, Cristian Canton Ferrer, and Zhengyou Zhang. Training deep networks for facial expression recognition with crowd-sourced label distribution, 2016. 1

[2] Qiong Cao, Li Shen, Weidi Xie, Omkar M Parkhi, and Andrew Zisserman. Vggface2: A dataset for recognising faces across pose and age. In *2018 13th IEEE international conference on automatic face & gesture recognition (FG 2018)*, pages 67–74. IEEE, 2018. 2

[3] Ian J. Goodfellow, Dumitru Erhan, Pierre Luc Carrier, Aaron Courville, Mehdi Mirza, Ben Hamner, Will Cukierski, Yichuan Tang, David Thaler, Dong-Hyun Lee, Yingbo Zhou, Chetan Ramaiah, Fangxiang Feng, Ruifan Li, Xiaojie Wang, Dimitris Athanasakis, John Shawe-Taylor, Maxim Milakov, John Park, Radu Ionescu, Marius Popescu, Cristian Grozea, James Bergstra, Jingjing Xie, Lukasz Romaszko, Bing Xu, Zhang Chuang, and Yoshua Bengio. Challenges in representation learning: A report on three machine learning contests, 2013. 1

[4] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition, 2015. 1

[5] Imagen-Team-Google, Jason Baldridge, Jakob Bauer, Mukul Bhutani, Nicole Brichtova, Andrew Bunner, Lluis Castrejon, Kelvin Chan, Yichang Chen, Sander Dieleman, Yuqing Du, Zach Eaton-Rosen, Hongliang Fei, Nando de Freitas, Yilin Gao, Evgeny Gladchenko, Sergio Gómez Colmenarejo, Mandy Guo, Alex Haig, Will Hawkins, Hexiang Hu, Huilian Huang, Tobenna Peter Igwe, Christos Kaplanis, Siavash Khodadadeh, Yelin Kim, Ksenia Konyushkova, Karol Langner, Eric Lau, Rory Lawton, Shixin Luo, Soňa Mokrá, Henna Nandwani, Yasumasa Onoe, Aäron van den Oord, Zarana Parekh, Jordi Pont-Tuset, Hang Qi, Rui Qian, Deepak Ramachandran, Poorva Rane, Abdullah Rashwan, Ali Razavi, Robert Riachi, Hansa Srinivasan, Srivatsan Srinivasan, Robin Strudel, Benigno Uria, Oliver Wang, Su Wang, Austin Waters, Chris Wolff, Auriel Wright, Zhisheng Xiao, Hao Xiong, Keyang Xu, Marc van Zee, Junlin Zhang, Katie Zhang, Wenlei Zhou, Konrad Zolna, Ola Aboubakar, Canfer Akbulut, Oscar Akerlund, Isabela Albuquerque, Nina Anderson, Marco Andreetto, Lora Aroyo, Ben Bariach, David Barker, Sherry Ben, Dana Berman, Courtney Biles, Irina Blok, Pankil Botadra, Jenny Brennan, Karla Brown, John Buckley, Rudy Bunel, Elie Bursztein, Christina Butterfield, Ben Caine, Viral Carpenter, Norman Casagrande, Ming-Wei Chang, Solomon Chang, Shamik Chaudhuri, Tony Chen, John Choi, Dmitry Churbanau, Nathan Clement, Matan Cohen, Forrester Cole, Mikhail Dektiarev, Vincent Du, Praneet Dutta, Tom Eccles, Ndidi Elue, Ashley Feden, Shlomi Fruchter, Frankie Garcia, Roopal Garg, Weina Ge, Ahmed Ghazy, Bryant Gipson, Andrew Goodman, Dawid Górny, Sven Gowal, Khyatti Gupta, Yoni Halpern, Yena Han, Susan Hao, Jamie Hayes, Jonathan Heek, Amir Hertz, Ed Hirst, Emiel Hoogeboom, Tingbo Hou, Heidi Howard, Mohamed Ibrahim, Dirichi Ike-Njoku, Joana Iljazi, Vlad Ionescu, William Isaac, Reena Jana, Gemma Jennings, Donovon Jenson, Xuhui Jia, Kerry Jones, Xi-

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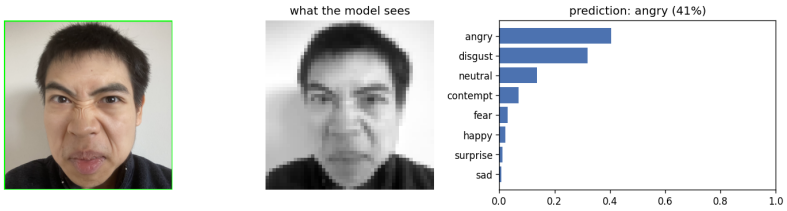
[6] Mehmet Emin Konuk and Erdal Kılıç. Efficientfer: Efficientnetv2 based deep learning approach for facial expression recognition. In *2025 7th International Congress on Human-Computer Interaction, Optimization and Robotic Applications (ICHORA)*, pages 1–7, 2025. 1, 3

[7] Alex Krizhevsky, Ilya Sutskever, and Geoffrey E Hinton. Imagenet classification with deep convolutional neural networks. In *Advances in Neural Information Processing Systems*. Curran Associates, Inc., 2012. 2

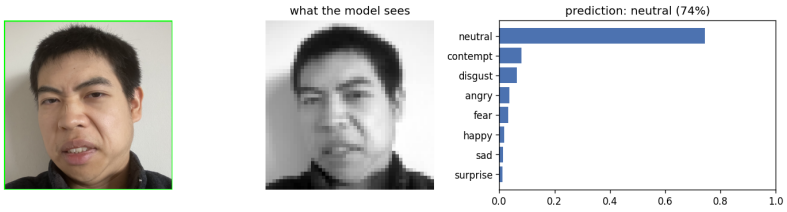
[8] Mingxing Tan and Quoc Le. Efficientnetv2: Smaller models and faster training. In *International conference on machine learning*, pages 10096–10106. PMLR, 2021. 1

[9] Sangdoo Yun, Dongyoon Han, Seong Joon Oh, Sanghyuk Chun, Junsuk Choe, and Youngjoon Yoo. Cutmix: Regularization strategy to train strong classifiers with localizable features, 2019. 1

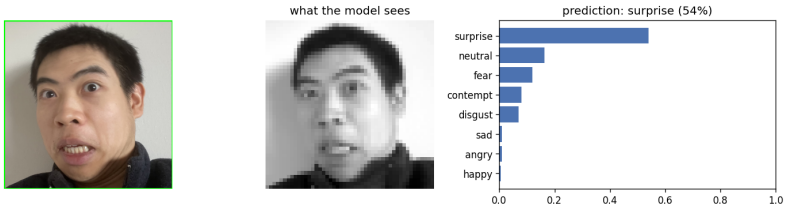
[10] Hongyi Zhang, Moustapha Cisse, Yann N. Dauphin, and David Lopez-Paz. mixup: Beyond empirical risk minimization, 2018. 1



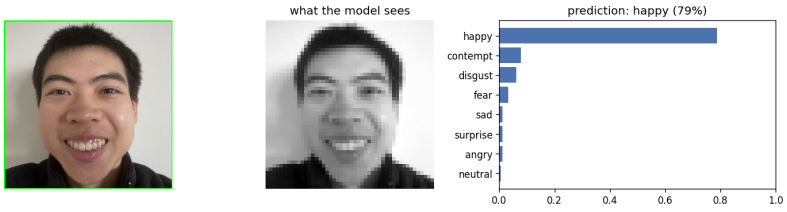
(a) Angry



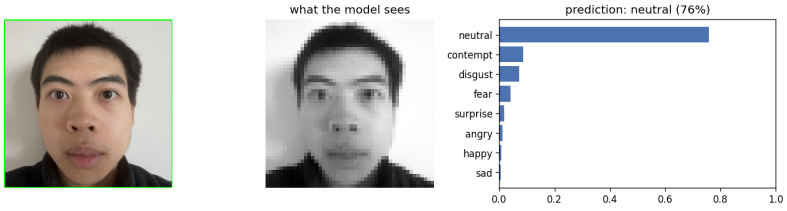
(b) Disgust



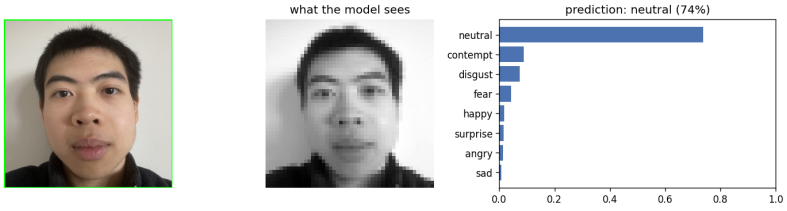
(c) Fear



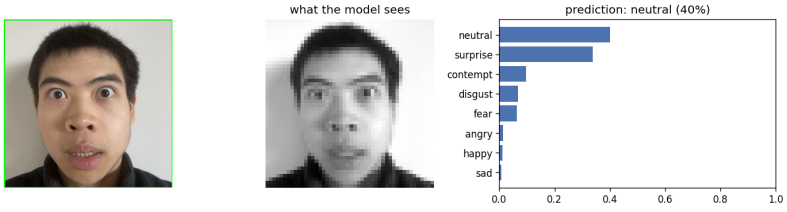
(d) Happy



(e) Neutral

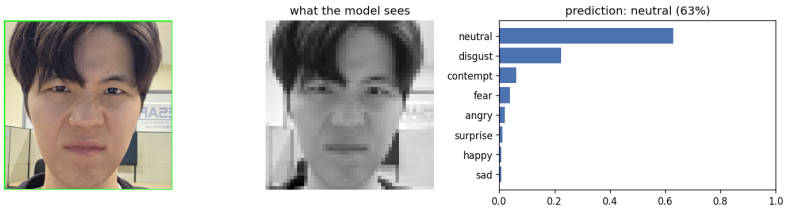


(f) Sad

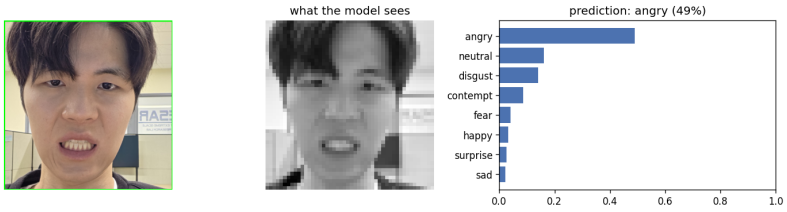


(g) Surprised

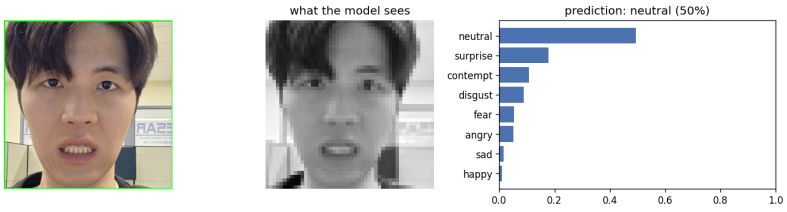
Figure 1. Random Stuart examples from the Stuart-Yujong dataset and results.



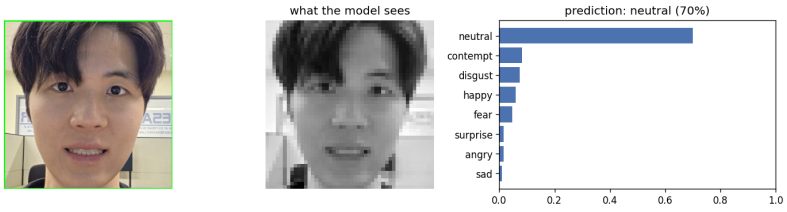
(a) Angry



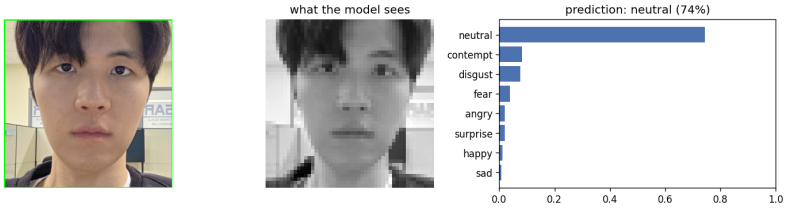
(b) Disgust



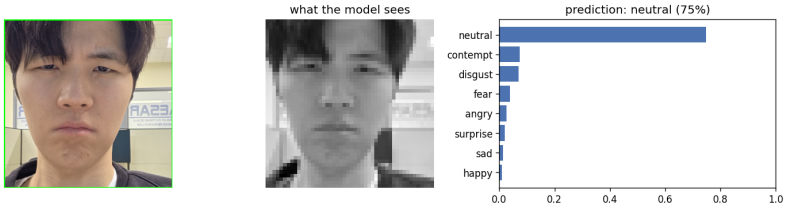
(c) Fear



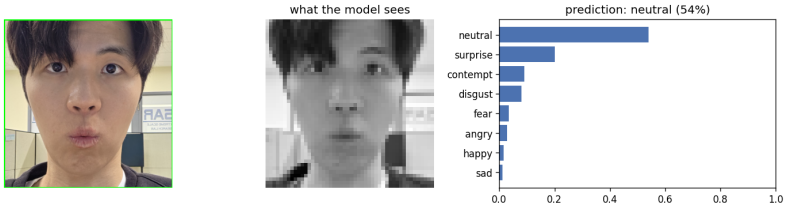
(d) Happy



(e) Neutral



(f) Sad



(g) Surprised

Figure 2. Random Yujong examples from the Stuart-Yujong dataset and results.